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UNIVERSITY OF GREENWICH
COMP1786 - Mobile
Application Design and
Development
Coursework

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COMP1786 – Mobile Application Design and Development Report

Table 1. Student Information

Your name	Lieu Vinh Khoi	Your Student ID	001391472
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1. SECTION 1 - BRIEF STATEMENT OF FEATURES YOU HAVE IMPLEMENTED

Table 2. Brief statement of features you have implemented

Feature	Status	Your Comments
Functionality A	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐ Not implemented ☐	I have fully implemented all the requirements for functionality A. In my application, users can enter the full details of a yoga class, including required fields like the day of the week, time, capacity, duration, price, and class type. I have also included an optional field for class descriptions, as specified. Additionally, a field like status information has been added to enhance functionality. Validation checks ensure all required fields are completed; otherwise, users receive an error prompt. Upon completion, the app displays these details for user confirmation, allowing them to review and modify entries before finalizing. This functionality has been thoroughly tested, and all specified features are working as expected.
Functionality B	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐ Not implemented ☐	I have fully implemented the features to store, view, edit, and delete yoga class details in an SQLite database. The `ClassListActivity` displays all saved classes, and allows users to filter by type, and search by fields. Each class can be edited or deleted with confirmation prompts for safe management. Additionally, I included database reset functionality, meeting all requirements specified in the task.
Functionality C	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐	I have successfully implemented all required features for managing class instances. The `EnterClassInstanceDetailsActivity` allows users to add,

	☐ Not implemented ☐	edit, and delete instances for specific yoga courses. I ensured that the instance date aligned with the yoga class's scheduled day. Additionally, users can assign an instructor and provide optional comments. All class instances are stored in SQLite, and users can select a class instance to update or delete it in `ClassInstanceListActivity`. These implementations fulfill the requirements outlined in the task.
Functionality D	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐ Not implemented ☐	I implemented a search feature that allows users to find classes by teacher name, supporting partial input to display all matching classes. Additionally, I added advanced options to search by `date` or `day of the week`, with a filter that enables users to refine results easily. Users can also select a class from the search results to view full details, fully meeting the requirements for this functionality.
Functionality E	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐ Not implemented ☐	I implemented a comprehensive data synchronization feature that allows users to upload all details of yoga classes to Firebase, my cloud-based database. My approach ensures that the app checks for network availability and uploads data only when Wi-Fi is connected. If Wi-Fi is unavailable, the app stores the data locally in SQLite and syncs it automatically when Wi-Fi becomes available or at periodic intervals. Additionally, if any data is updated locally, these changes are marked with a status field and subsequently synchronized with Firebase upon reconnection. This setup ensures data consistency between the local and cloud databases, fully meeting the requirements for this functionality.
Functionality F	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐ Not implemented ☐	I extended the core requirements by adding several features that improve user management and accessibility in the admin app. I implemented user authentication to secure access to the admin section, ensuring only authorized users can manage class and user information. Additionally, I introduced role-based access, where each user is assigned a role (e.g., admin,

		customer, or teacher), allowing specific views and permissions based on their role for each app. Teachers, for instance, can view only the class instances assigned to them. To enhance data handling, I added filter spinners on various pages to streamline data retrieval, making the interface more intuitive and user-friendly.
Functionality G	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐ Not implemented ☐	I developed a hybrid app using .NET MAUI specifically for customers interested in booking yoga classes. This app is fully integrated with Firebase, providing real-time access to yoga classes and instances created by the Android Admin app. Customers can register, log in, browse available classes, view detailed information, and select specific class instances to add to their cart for booking. Additionally, I implemented a search function, allowing users to filter classes by day of the week or time of day, which enhances the user experience by helping them find sessions that best fit their schedules. This approach fulfills all requirements outlined for the hybrid customer app.
Functionality H	Fully completed ☒ Partially completed ☐ Having bugs/Not working ☐ Not implemented ☐	I implemented the booking feature in the hybrid app, allowing users to add class instances to a shopping cart and then submit it to Firebase with their ID for record-keeping. The booking model captures key details, including a list of selected class instances, total price, and user information, ensuring that each booking is associated with the correct user. Additionally, users can view all their booked classes in one place, streamlining their experience by providing access to all booked in the app. This setup meets all outlined requirements for the customer booking functionality.

Link to recorded video (if you record your application before submitting the report)

The recorded video will demonstrate the implemented product. It is roughly 10 minutes.

2. SECTION 2 - REFLECTION

2.1 Reflections on Android and Cross-Platform App Development

Working on the Universal Yoga project, where I developed both an Android admin app and a cross-platform customer app, was an in-depth journey filled with valuable lessons in app architecture, user security, data synchronization, and UI design.

I started with the admin app, prioritizing secure user authentication by implementing SHA-256 password hashing with salt to keep data protected. Setting up a local SQLite database was a must for offline functionality, and syncing it with Firebase whenever Wi-Fi became available became both a challenge and a deep learning experience. I researched different techniques for efficient data sync, ultimately setting up periodic and connectivity-based synchronization to keep the data updated automatically.

A significant milestone was syncing data from Firebase to SQLite, particularly on the first app launch, which gave me insight into Android database handling. With the data flow solidified, I built out essential features for managing yoga classes, bookings, and user accounts. Creating a functional, user-friendly UI was challenging but rewarding, requiring attention to usability and exhaustive testing, especially around critical actions like adding, updating, or deleting items.

Moving on to the customer app, which I built in .NET MAUI on Visual Studio 2022, I could rely on the sync structure I'd already set up. This app skipped the local database, syncing directly with Firebase for real-time updates and simplifying data handling. My focus here was to create a smooth experience, allowing users to browse and book classes effortlessly. Designing an engaging UI was a challenge—features like a “last booking list” and highlights for new classes made the app intuitive and enjoyable to use.

This project was a major learning experience, giving me hands-on skills with Firebase integration, secure data management, structured error handling, and UI design. Each obstacle pushed me to think creatively and build a stronger app. Looking back, I'm grateful for the challenges; they ultimately made the app better and deepened my understanding, giving me skills I'll carry forward into future projects.

2.2 Usecase

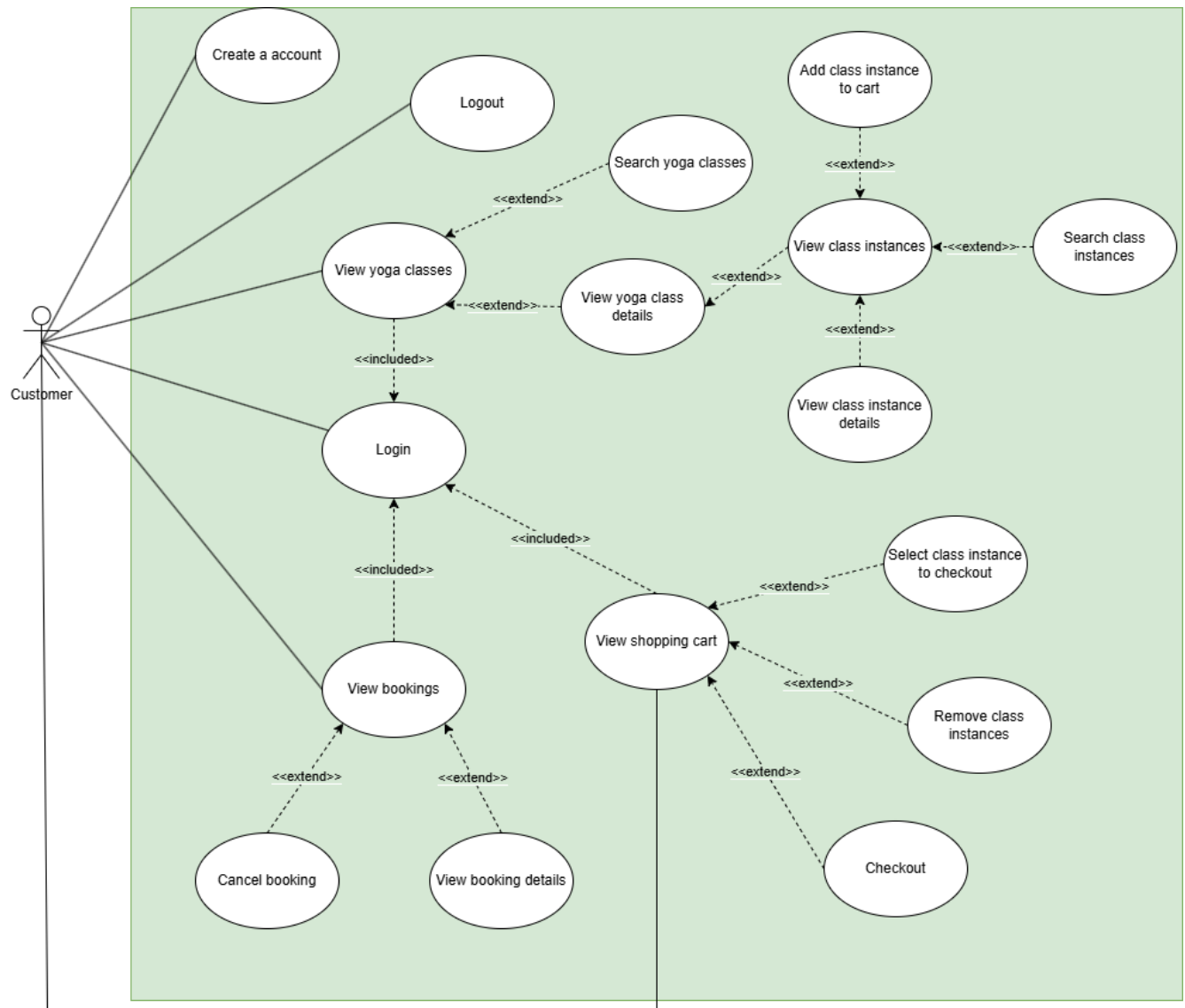


Figure 1. Use case of customer role

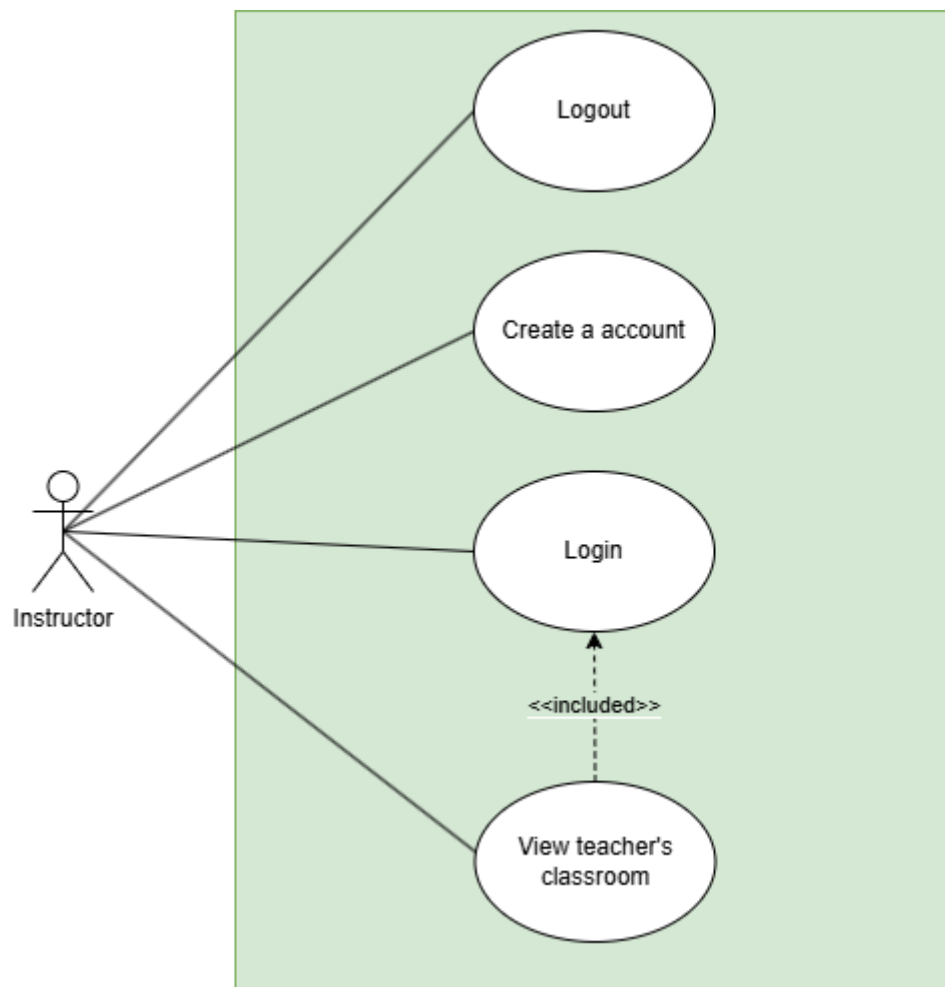


Figure 2. Usecase of instructor role



Figure 3. General usecase of the Admin role

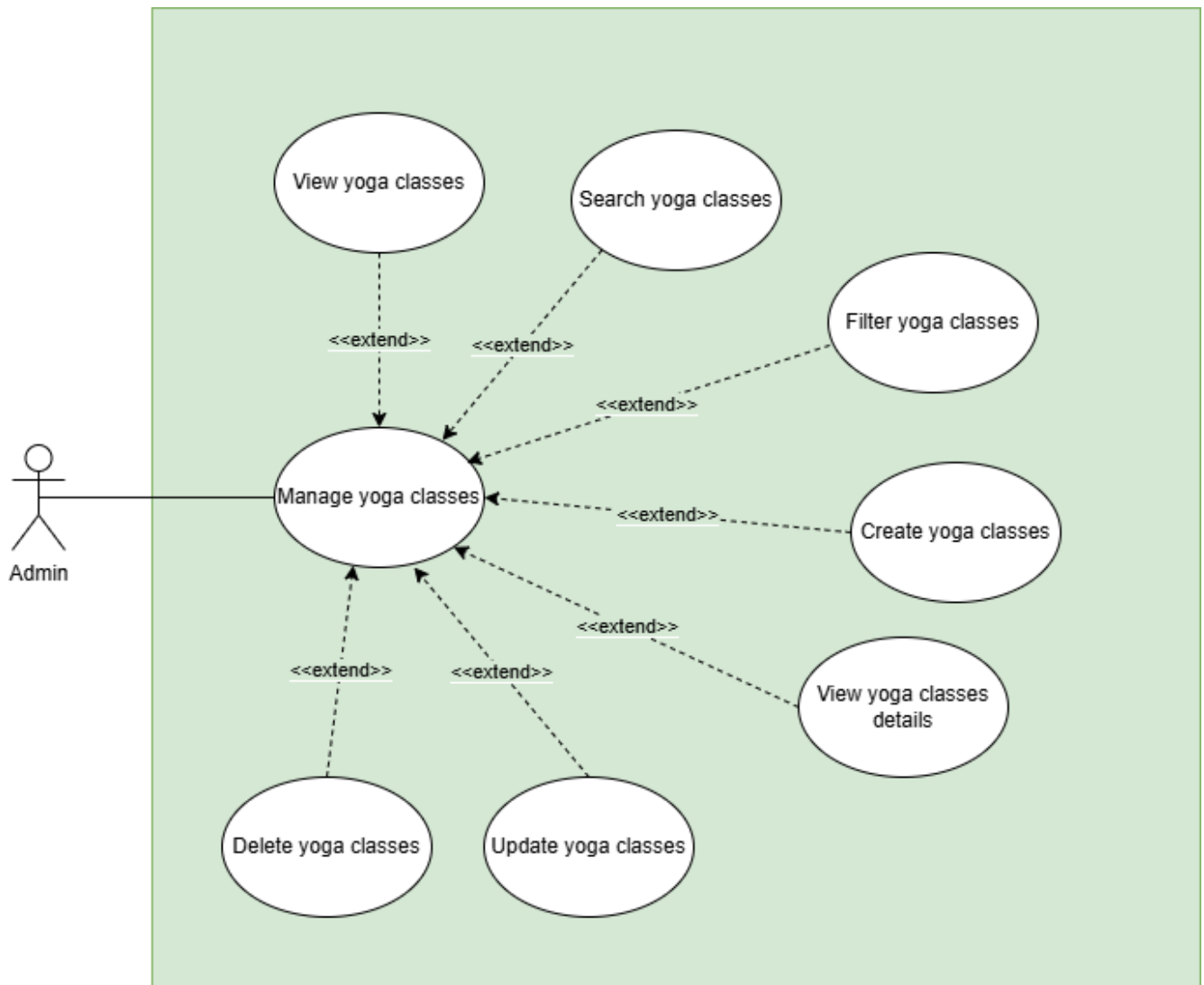


Figure 4. Detailed Use Case Diagram for Admin Yoga Class Management Functions

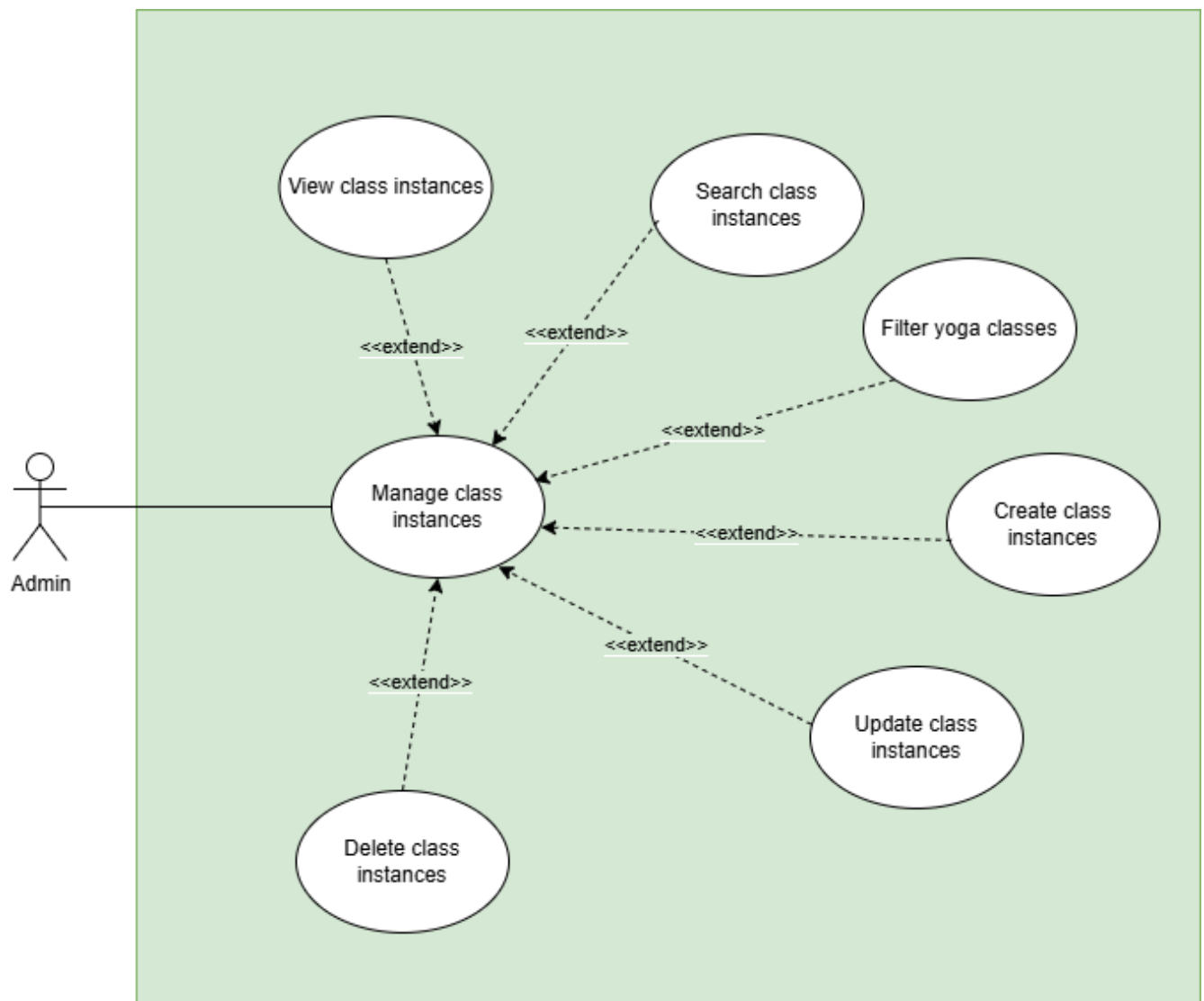


Figure 5. Detailed Use Case Diagram for Admin Class Instance Management Functions

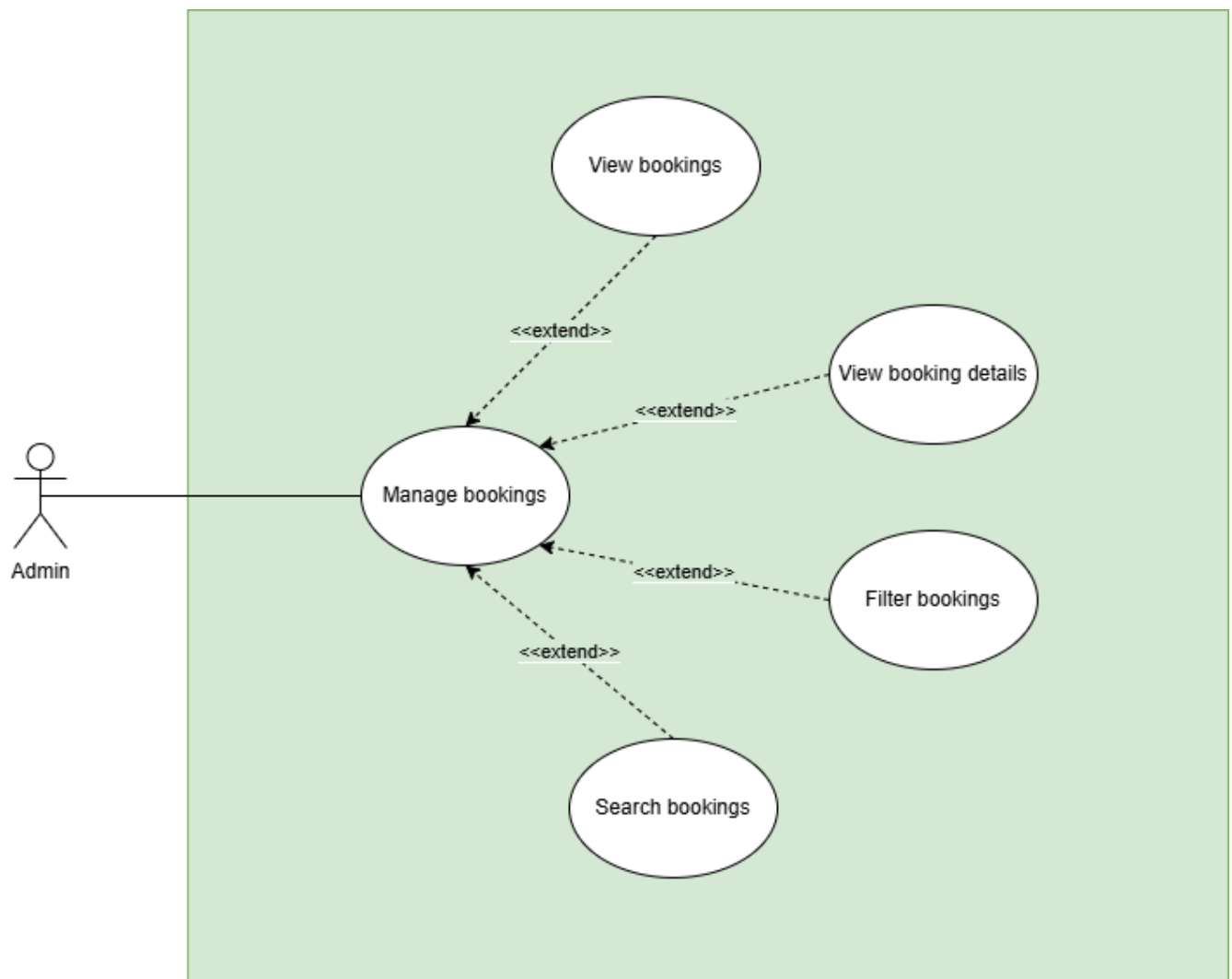


Figure 6. Detailed Use Case Diagram for Admin Booking Management Functions

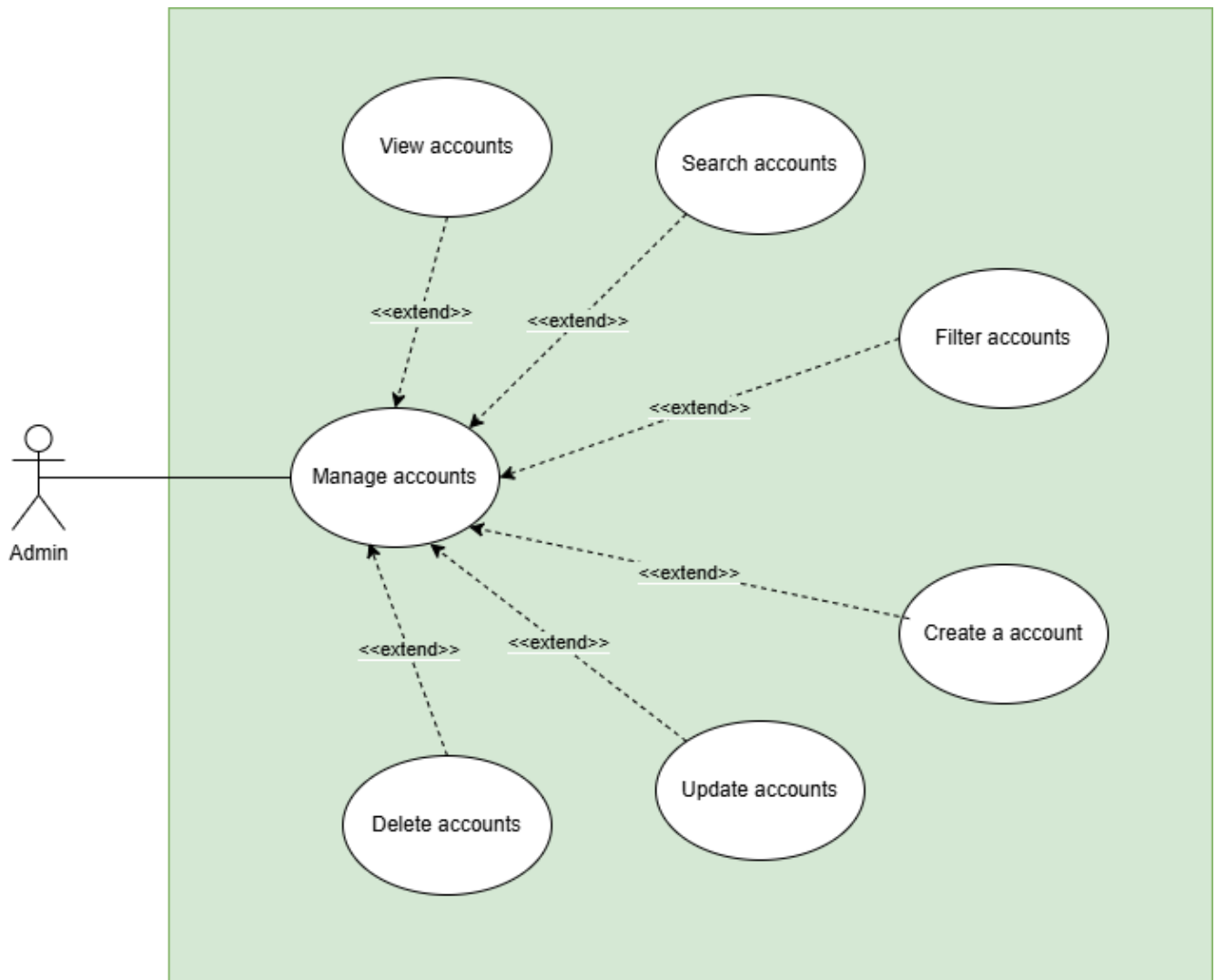


Figure 7. Detailed Use Case Diagram for Admin Account Management Functions

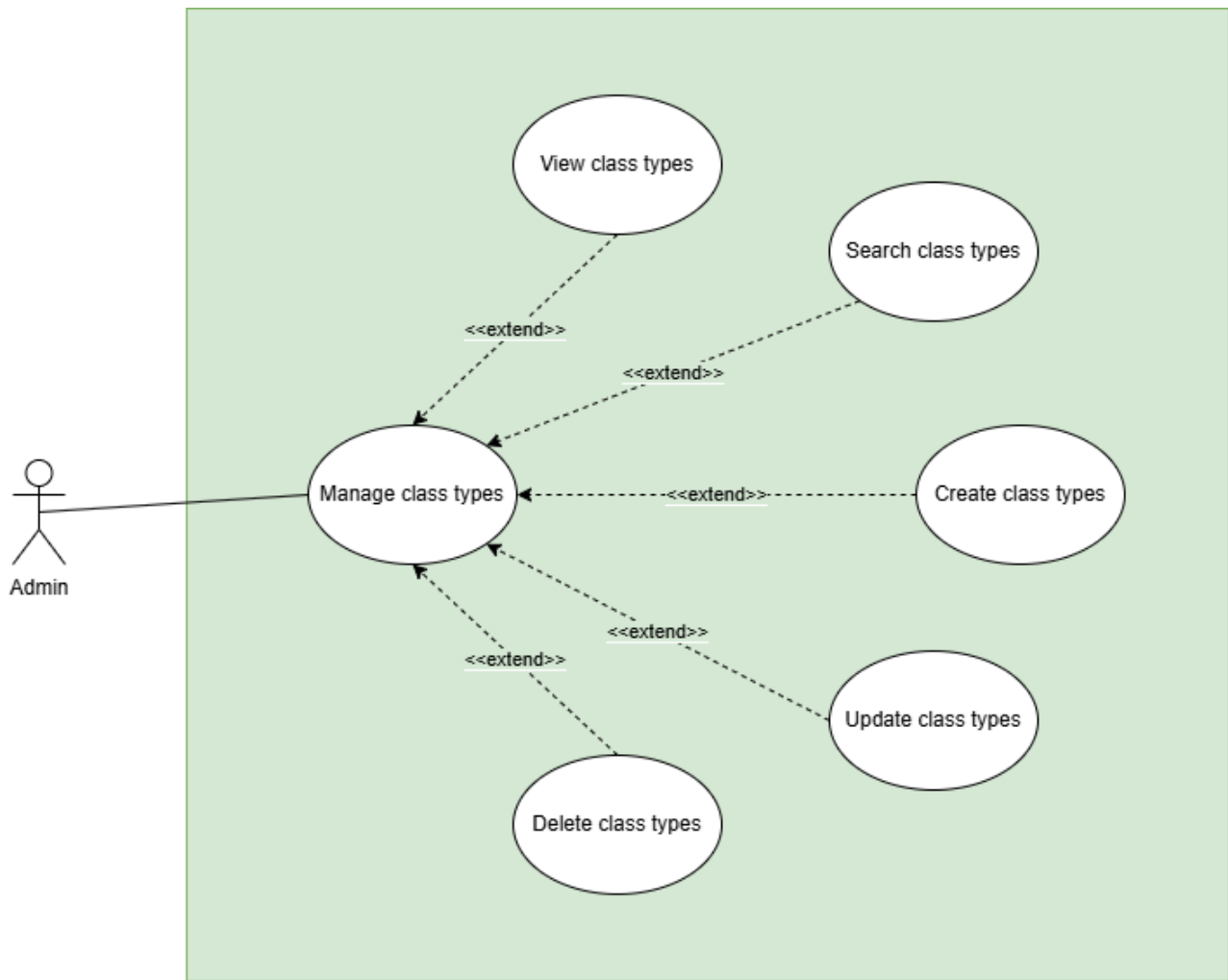


Figure 8. Detailed Use Case Diagram for Admin Class Type Management Functions

2.3 Entity-Relationship Diagram

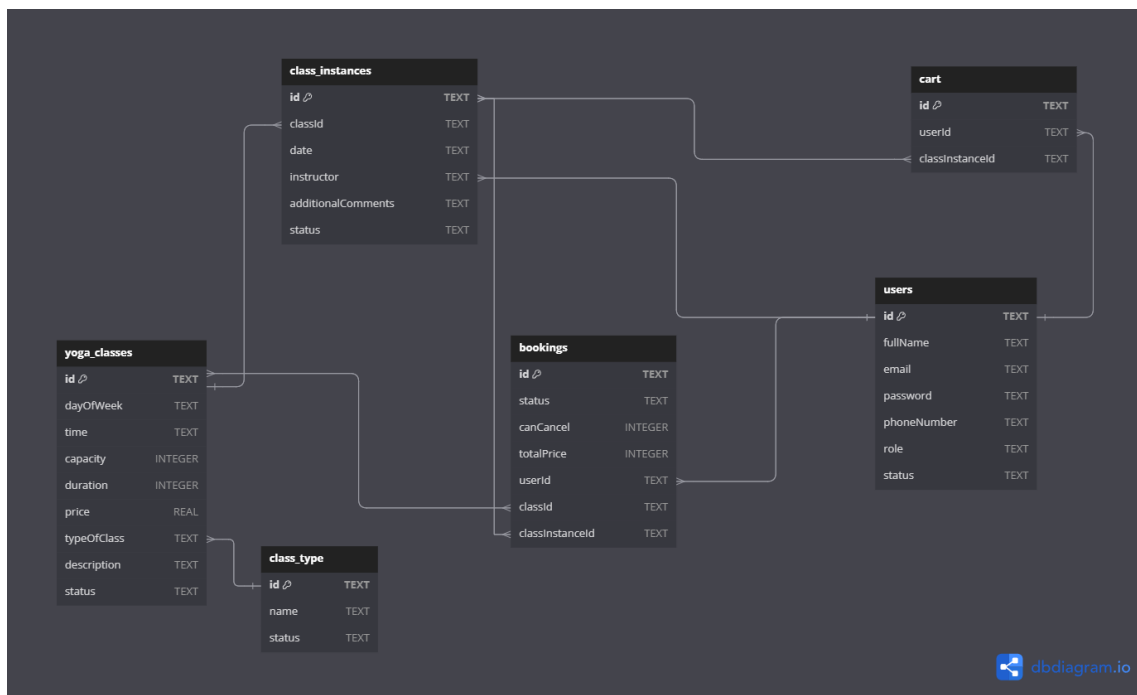


Figure 9. Entity-Relationship Diagram

2.4 Wireframe

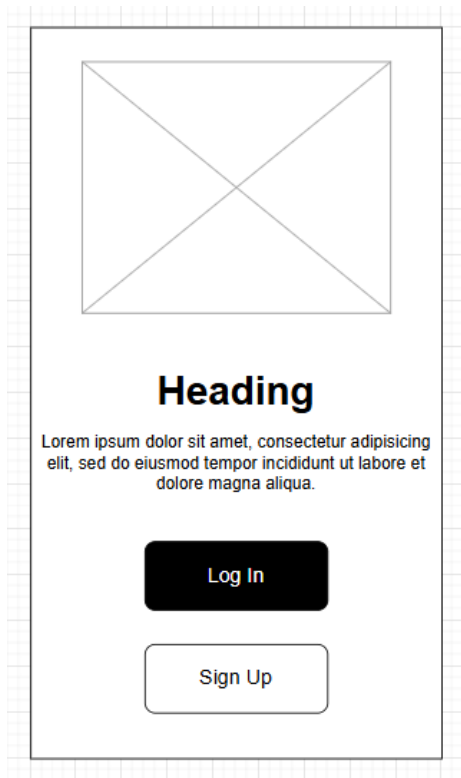


Figure 10. Welcome page

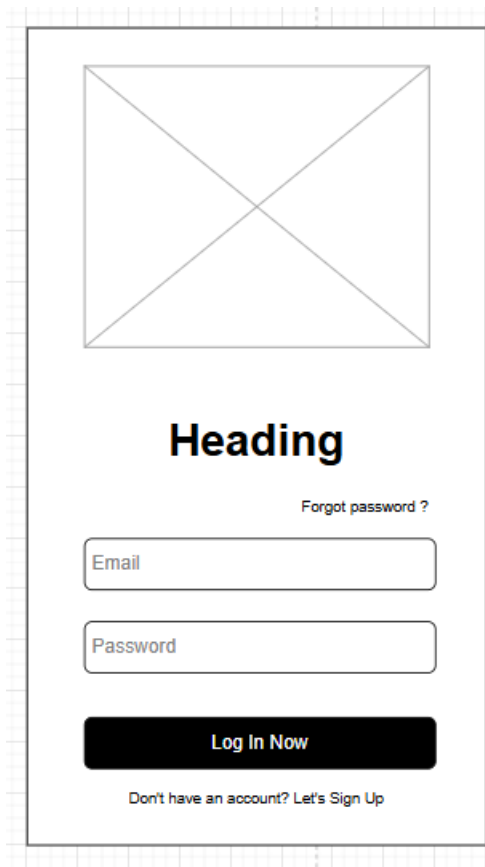
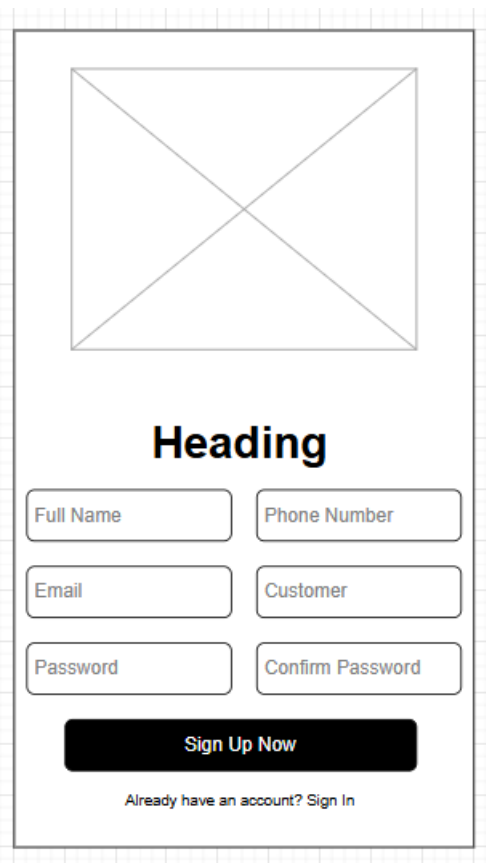
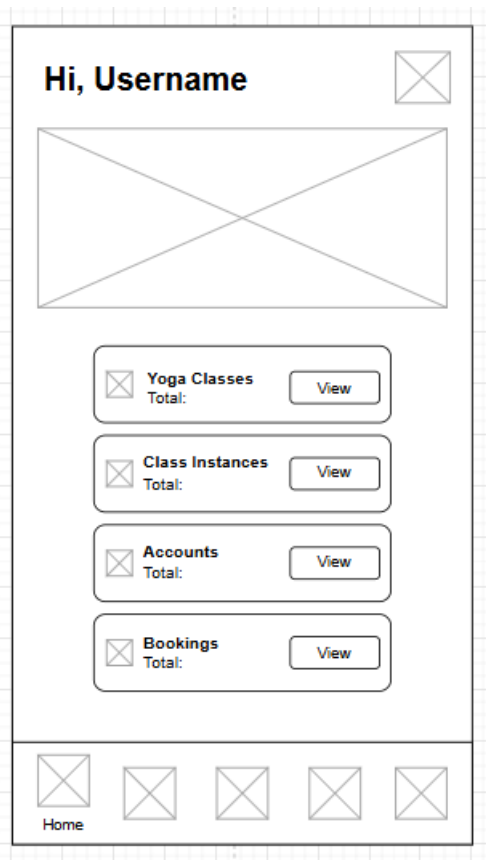


Figure 11. Login page



A wireframe of a signup page. At the top is a large square placeholder with an 'X' inside. Below it is a heading. The form consists of six input fields arranged in two columns: 'Full Name', 'Phone Number', 'Email', 'Customer', 'Password', and 'Confirm Password'. A black 'Sign Up Now' button is centered below the fields. At the bottom, there is a link that says 'Already have an account? Sign In'.

Figure 12. Signup page



A wireframe of an admin homepage. At the top left is a greeting 'Hi, Username' followed by a small square placeholder with an 'X'. Below this is a large rectangular placeholder with an 'X'. The main content area contains four summary cards, each with a small square placeholder with an 'X', a title, a 'Total:' label, and a 'View' button. The cards are for 'Yoga Classes', 'Class Instances', 'Accounts', and 'Bookings'. At the bottom is a navigation bar with five square placeholders with 'X's; the first one is labeled 'Home'.

Figure 13. Homepage of Admin

☐
Class

+ Add Class

Search for...

All ☐

☐ Type of class
 ☐ Price

☐ Day of week
 ☐ Time
 ☐ Duration

☐ Description
 ☐ Status
 ☐

☐ Type of class
 ☐ Price

☐ Day of week
 ☐ Time
 ☐ Duration

☐ Description
 ☐ Status
 ☐

☐ Type of class
 ☐ Price

☐ Day of week
 ☐ Time
 ☐ Duration

☐ Description
 ☐ Status
 ☐

☐
☐
☐
☐
☐

Class

Figure 14. Yoga class management

☐
Add class

Wifi:

Day of the Week
 Monday ☐

Time

00

45

Capacity
 Capacity

Duration (minutes)

45

Price
 Price (\$)

Type of Class
 Flow Yoga ☐

Description (Optional)
 Description (Optional)

Save

Figure 15. Add yoga class

Class Instances

+ Add Instance

Date

Instructor

Type of class - Day of week - Time

Status

Description

Date

Instructor

Type of class - Day of week - Time

Status

Description

Date

Instructor

Type of class - Day of week - Time

Status

Description

Class Instance

Figure 16. Class instance management

Class Instance

Wifi:

Select Yoga Class

Flow Yoga - Monday - (10:00 - 11:00)

Date

Select Date

Instructor

Carla Smith

Additional Comments

Enter Additional Comments

Save Class Instance

Figure 17. Add class instance

☐

Bookings

All ☐

☐ Full name
 ☐

☐ Total Price
 ☐ Status

☐ Yoga Classes
 ☐ Class

☐ Full name
 ☐

☐ Total Price
 ☐ Status

☐ Yoga Classes
 ☐ Class

☐ Full name
 ☐

☐ Total Price
 ☐ Status

☐ Yoga Classes
 ☐ Class

☐

☐

☐

☐ Booking

☐

Figure 18. Bookings management

☐

Booking Details

☐ Full Name:

☐ Total Price:

☐ Status:

Yoga Classes

Type of Class - Day of Week (Time)

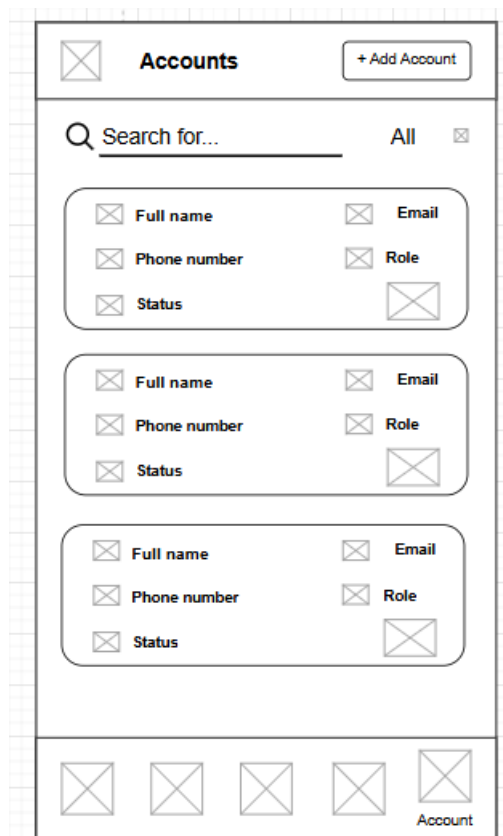
☐ Date - ☐ Instructor:

☐ Date - ☐ Instructor:

Confirm

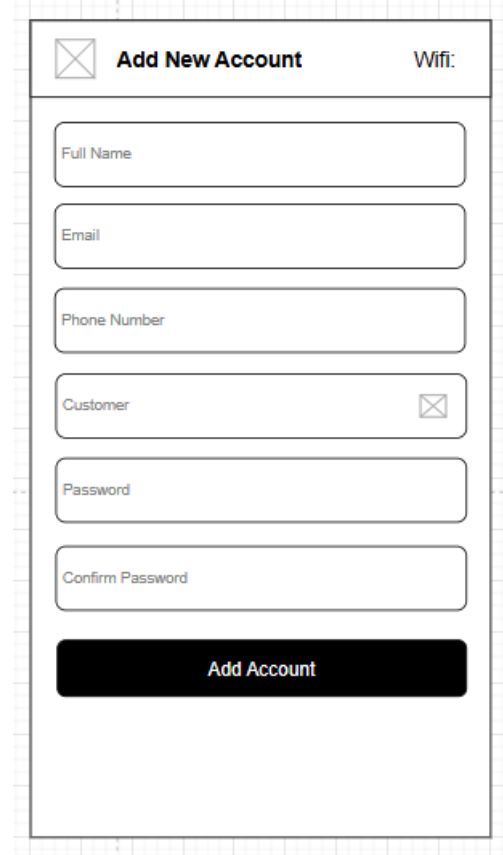
Reject

Figure 19. Booking details



The 'Accounts' screen features a top header with a back arrow, the title 'Accounts', and a '+ Add Account' button. Below the header is a search bar with a magnifying glass icon and the placeholder text 'Search for...'. To the right of the search bar is a filter dropdown menu currently set to 'All'. The main content area displays three identical account cards. Each card contains a grid of fields: 'Full name', 'Email', 'Phone number', 'Role', and 'Status', each with a checkbox icon. At the bottom of the screen, there is a horizontal list of five account icons, with the last one labeled 'Account'.

Figure 20. Account management



The 'Add New Account' screen has a top header with a back arrow, the title 'Add New Account', and a 'Wifi:' label. The form consists of several input fields: 'Full Name', 'Email', 'Phone Number', 'Customer' (with a checkbox icon), 'Password', and 'Confirm Password'. At the bottom of the form is a prominent black button with the text 'Add Account' in white.

Figure 21. Add a new account

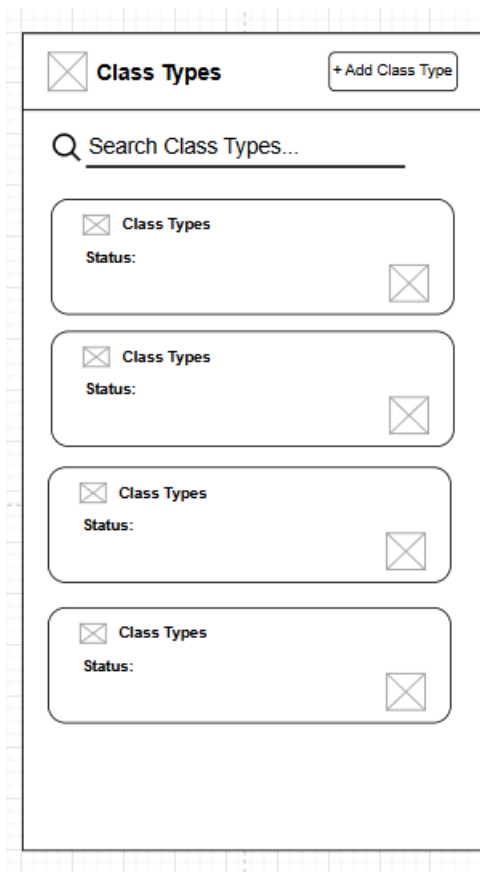


Figure 22. Class type management

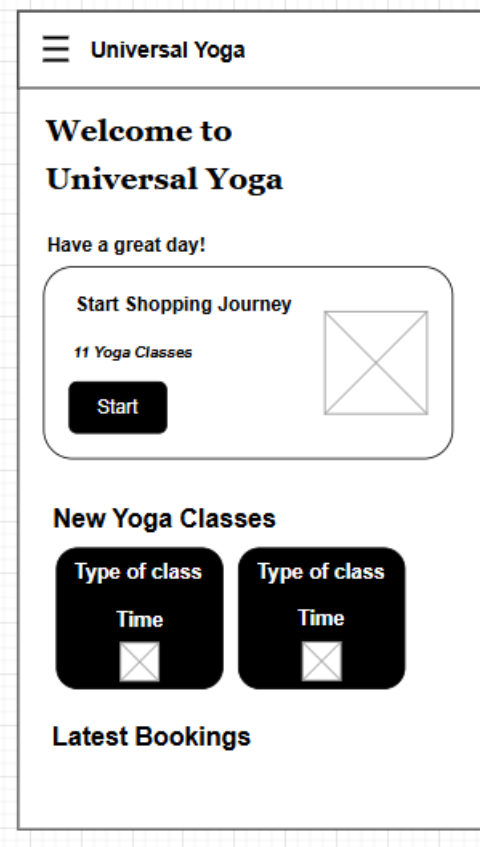


Figure 23. Homepage of Customer

Yoga Class List

Choose a search option:

☒ Search by Day of the Week

☐ Search by Time of Day

☒ Monday

☒ Tuesday

☒ Wednesday

Type of Class

Day of week

Time

Type of Class

Day of week

Time

Figure 24. Yoga class list with search by day of week

Yoga Class List

Choose a search option:

☐ Search by Day of the Week

☒ Search by Time of Day

Enter time of day

Type of Class

Day of week

Time

Type of Class

Day of week

Time

Figure 25. Yoga class list with search by time of day

< Yoga Class Details

Type of class

Description:

Price:

Time:

Capacity:

Day:

Book Yoga Class

Figure 26. Yoga class details

< Book Yoga Class

Select the course you want to join:

Search by date

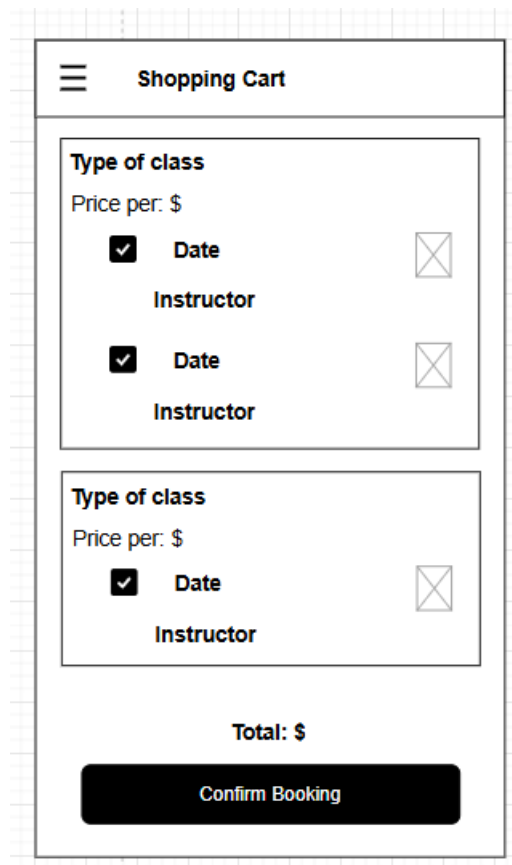
Date: ☒ Instructor: Comments:

Date: ☒ Instructor: Comments:

Total: \$

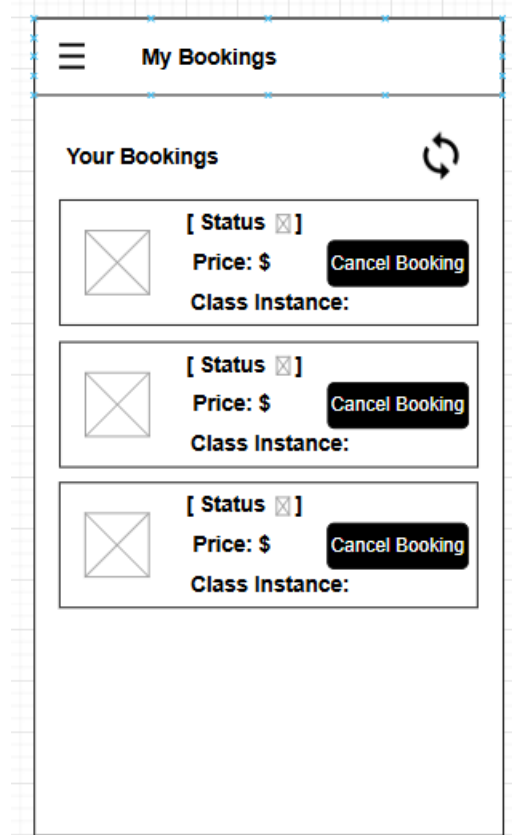
Add to Cart

Figure 27. Book yoga class



The 'Shopping Cart' screen features a hamburger menu icon on the left. The main content area contains two identical class selection boxes. Each box is titled 'Type of class' and includes a 'Price per: \$' label. Inside each box, there are two rows of options: a checked checkbox next to 'Date' and an unchecked checkbox next to 'Instructor'. Below these boxes, the text 'Total: \$' is displayed, followed by a large black button labeled 'Confirm Booking'.

Figure 28. Shopping cart



The 'My Bookings' screen features a hamburger menu icon on the left. The main content area is titled 'Your Bookings' and includes a refresh icon. Below this title, there are three identical booking entries. Each entry consists of a placeholder icon (a square with an 'X'), a status label '[Status]' with a refresh icon, a 'Price: \$' label, and a 'Class Instance:' label. To the right of each entry is a black button labeled 'Cancel Booking'.

Figure 29. My Bookings